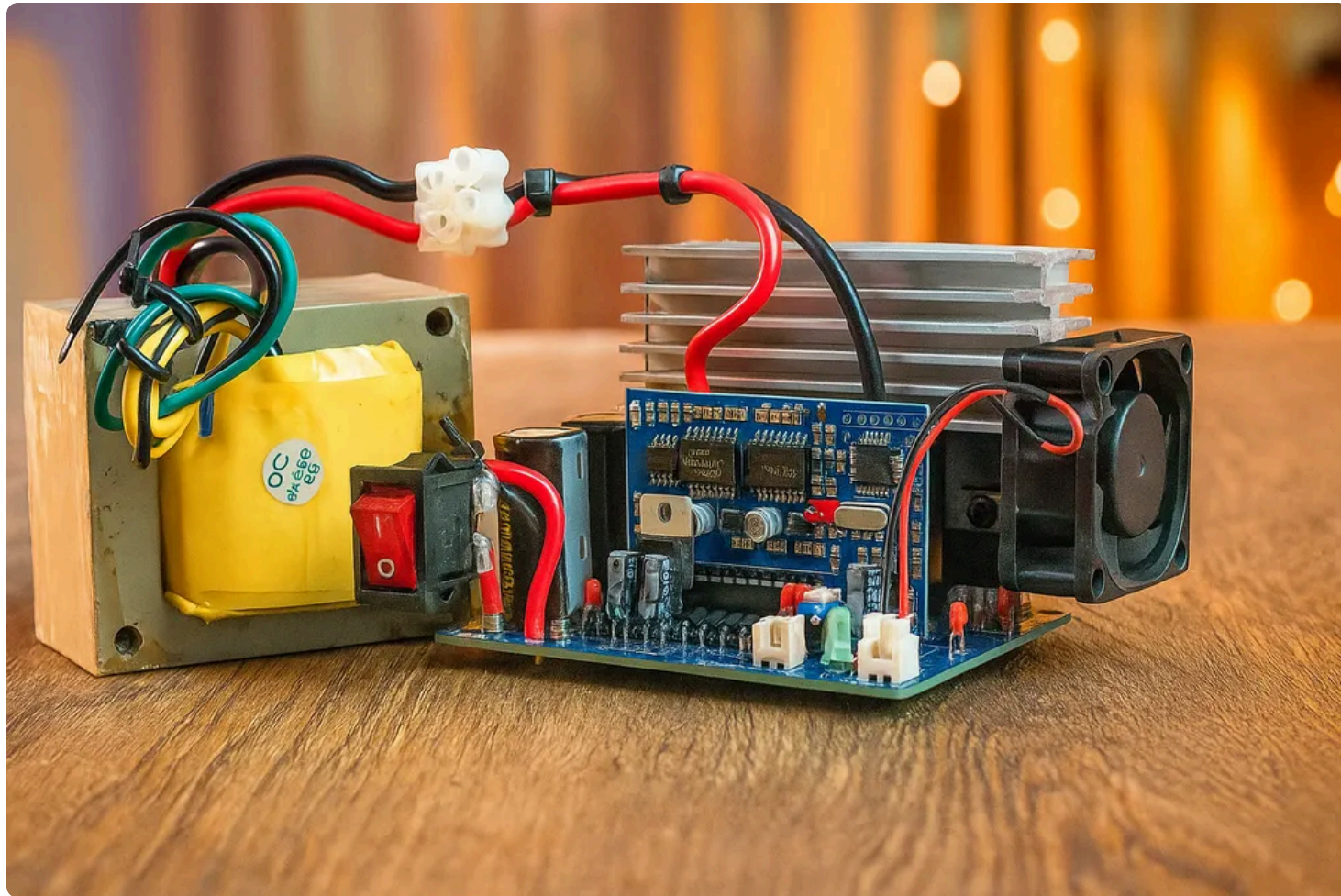


How to Make a Pure Sine Wave Inverter Using the EGS002 Module and UPS Transformer

By Jhuman in Circuits > Electronics  7,069  2



Learn how to build a Pure Sine Wave Inverter using an EGS002 module and a UPS Transformer. Get the complete circuit diagram, wiring instructions, and working explanation.

A Pure Sine Wave Inverter is ideal for providing clean, stable AC power that mimics your mains supply. This is crucial for powering sensitive electronics like medical equipment, TVs, and PCs. In this guide, we will walk you through making a Pure Sine Wave Inverter using the highly popular **EGS002 module**, a **UPS Transformer**, and other components.

Supplies

Main Components You Will Need

Here's the list of primary components required:

1. **EGS002 Pure Sine Wave Inverter Module** – The brain of the inverter.
2. **MOSFETs (4–6 units):** IRF3205 or IRFZ44N MOSFETs for switching.
3. **Transformer:** A UPS-style 12–0–12V or 24–0–24V (based on your input voltage).
4. **Filter Capacitors:** 4700 μ F, 100 μ F, and 10 μ F for smoothing and filtering.
5. **Resistors (R1–R13):** Gate resistors, feedback resistors, and sense resistors.
6. **Diodes (D1–D4):** FR207 fast recovery diodes for protection and feedback.
7. **Fan Transistor:** S8050 or any NPN transistor for fan drive.
8. **Voltage Regulator:** LM7805 for providing a stable 5V supply.
9. **Additional Parts:** Terminal connectors, PCB, heatsinks, and cooling fan.

Step 1: ⚡ What Is a Pure Sine Wave Inverter?

A **Pure Sine Wave Inverter** is a device that **converts DC (Direct Current)** from a battery into **clean AC (Alternating Current)** with a smooth sinusoidal waveform, just like the power you get from the grid. It is ideal for running sensitive electronics (computers, TVs, medical equipment), because its output mimics the quality and stability of the utility grid.

✅ The Fundamental Concept:

1. **Input Source (Battery):**
2. The process begins with a low-voltage **DC source** — typically a 12V, 24V, or 48V battery.
3. **SPWM (Sinusoidal Pulse Width Modulation) Generation:**
4. An **SPWM Controller** (such as an EGS002, SG3525, or a microcontroller) generates a series of rapid ON/OFF pulses.
 - These pulses have a **duty cycle** that varies smoothly like a sine wave.
 - This means the pulse widths are longer when the sine wave is at its peak and narrower when it is closer to zero.
5. **Power Switching (MOSFETs or IGBTs):**

6. The SPWM signals drive MOSFETs or IGBTs in an **H-Bridge** configuration.
7. This H-Bridge alternately applies the battery voltage across the transformer's primary winding in a manner that mimics the shape of an AC sine wave.
8. **Step-Up Transformer:**
9. The H-Bridge applies the SPWM signals to the primary winding of the **power transformer** (often called the inverter or UPS transformer).
10. The transformer steps the voltage from 12V/24V DC (converted into SPWM pulses) to 230V or 120V AC (depending on design).
11. **Output Filtering:**
12. An **LC filter** (inductor and capacitor) at the output removes the sharp edges of the SPWM pulses, smoothing them out into a clean, low-distortion **pure sine wave**.

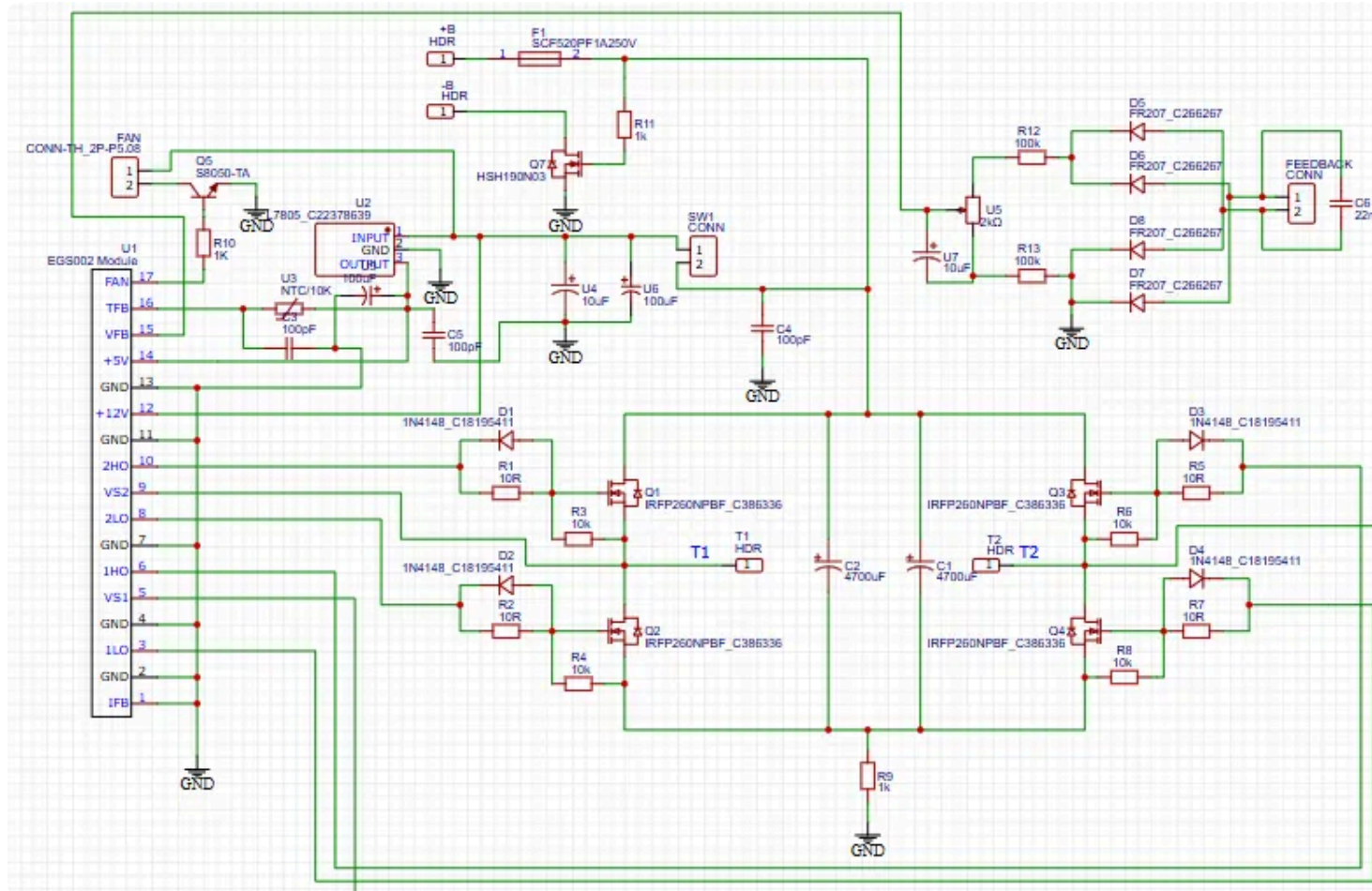
Step 2: What Is the EGS002 Module?

The **EGS002** is a dedicated Pure Sine Wave Inverter Controller. It's a compact, intelligent device that:

1. Generates SPWM (Sinusoidal Pulse Width Modulation) signals.
2. Drives MOSFETs for switching.
3. Provides output voltage feedback and protection.
4. Enables features like soft-start, overload protection, and low-battery protection.

With an EGS002, making a sine wave inverter becomes much simpler compared to using microcontroller-based or analog PWM designs.

Step 3: Circuit Diagram



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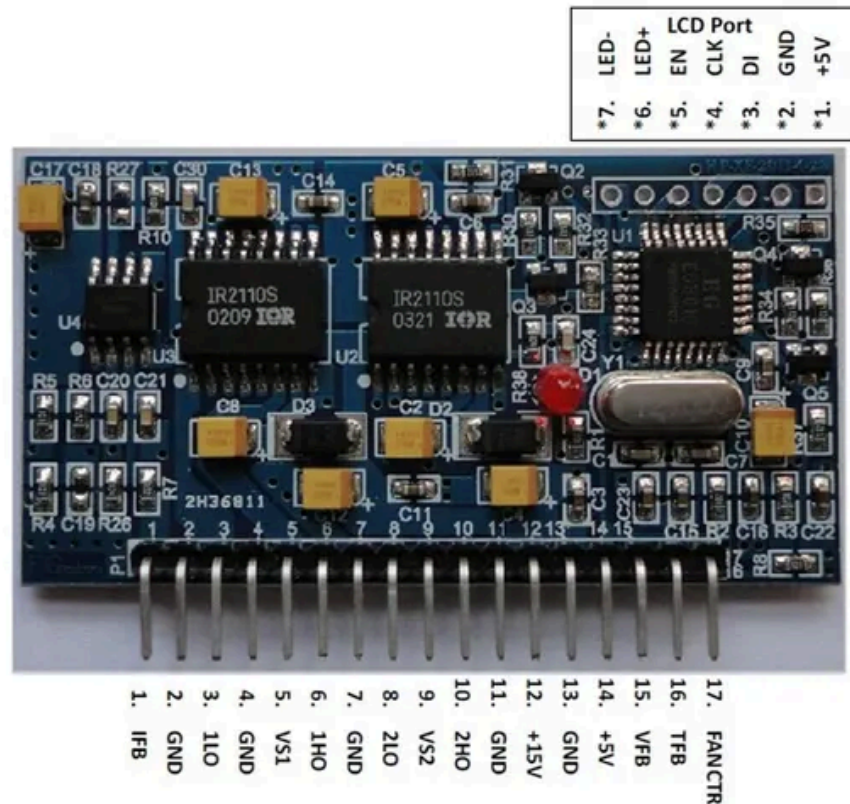
The inverter uses an EGS002 module, four MOSFETs (IRF3205 or IRFZ44N), a UPS Transformer, a 12V or 24V battery, and supporting components. The circuit connects these together to produce a stable AC output.

Circuit Diagram Explanation (Serial by Serial):

1. The EGS002 module is at the heart of the design. Its SPWM outputs (H0 and L0) drive the gates of four MOSFETs arranged in an H-Bridge configuration.
2. The H-Bridge alternately applies the 12V or 24V supply across the primary winding of the **UPS Transformer**, creating a sine wave voltage at its output.
3. The **feedback circuit** (using diodes and resistors) connects the output voltage back to the EGS002 module, allowing it to adjust its PWM and maintain a stable output voltage.
4. The **fan transistor (Q5)** is driven by the EGS002 module. It turns ON the cooling fan when needed (according to the module's thermal and load monitoring).
5. The filter capacitor (C1) across the supply lines smoothens the DC supply. The smaller capacitors (C4, C5, C6) reduce noise and stabilize the module.

6. The MOSFET gate resistors (R1–R8) limit the gate charging and discharging currents, providing better switching characteristics and longer MOSFET life.
7. The F1 fuse provides overcurrent protection for the circuit.
8. The LM7805 regulator provides a stable +5V supply for the EGS002 module.

Step 4: Pin Connections of the EGS002

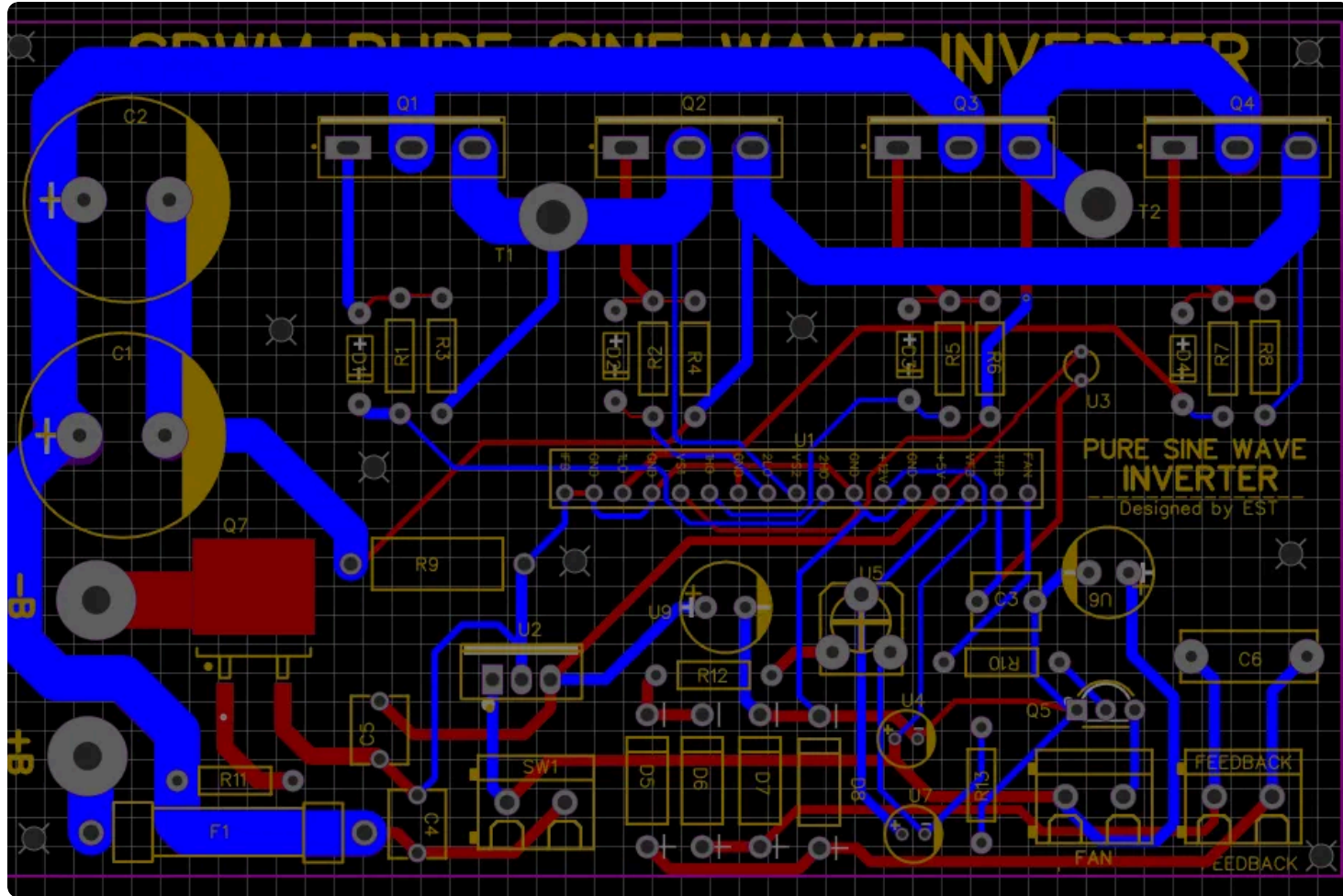


Here's how you connect the EGS002 in this design:

1. **Pin 1 & 2 (GND):** Ground connections.

2. **Pin 3 (IFB):** Current feedback from the output to regulate load and maintain stable voltage.
3. **Pin 5 (L0):** Low Side Gate Drive for MOSFETs.
4. **Pin 7 (H0):** High Side Gate Drive for MOSFETs.
5. **Pin 9 & 10 (VS1, VS2):** Voltage sense inputs for output voltage feedback.
6. **Pin 17 (FAN):** Output for controlling the cooling fan.
7. **Additional Connections:** The EGS002 requires a +5V and +12V supply for internal logic and gate driving.

Step 5: How the Circuit Works



Here's the working flow:

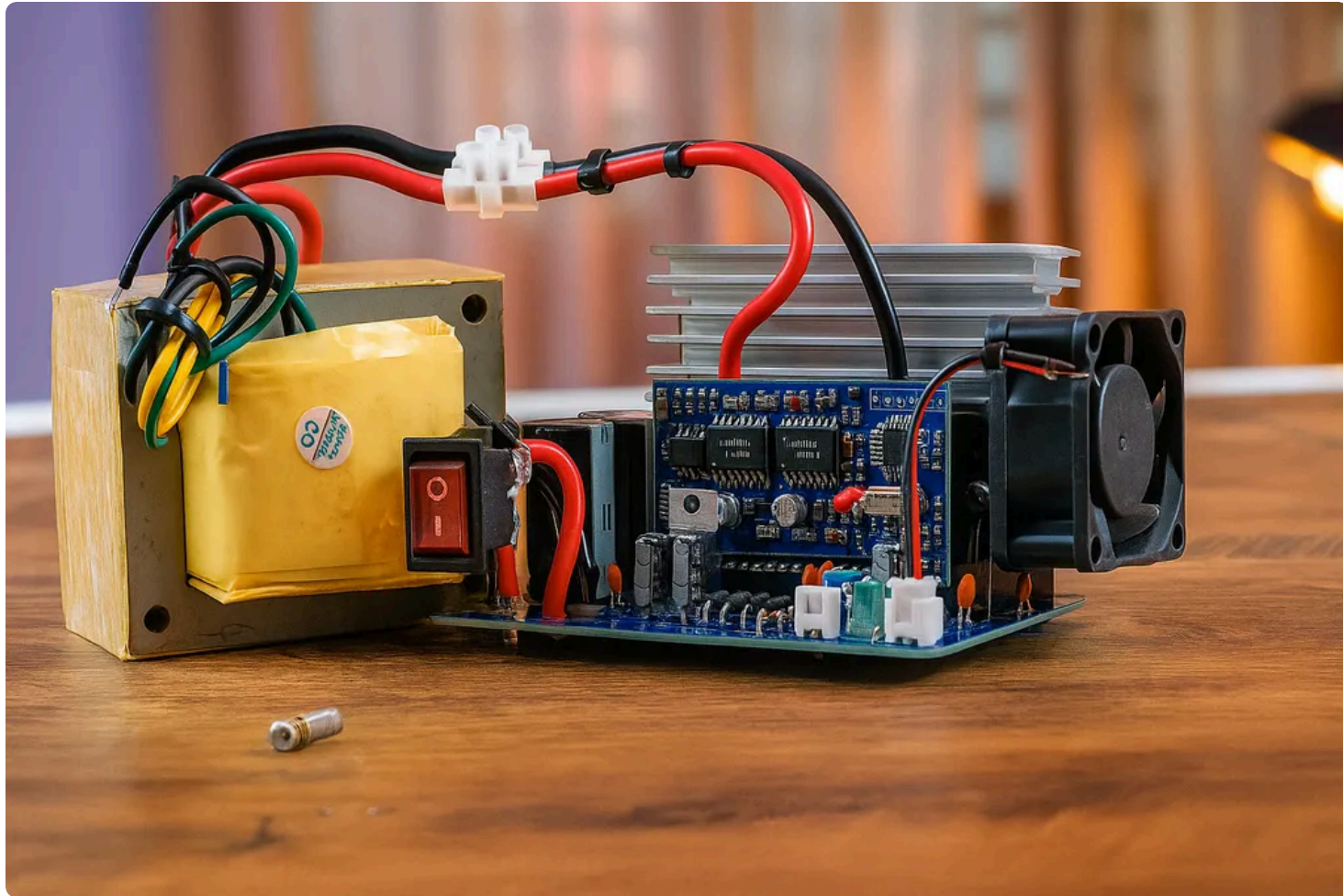
1. The EGS002 generates SPWM signals based on its internal clock and feedback.
2. The MOSFET drivers apply these SPWM signals to the gates of the MOSFETs.
3. The MOSFETs drive the UPS Transformer's primary winding, producing an AC voltage across the secondary.
4. The feedback network monitors the output voltage and adjusts the PWM accordingly, ensuring a stable output regardless of load variations.
5. The output can be connected to any load such as a bulb, fan, or sensitive electronics.

Step 6: Testing and Final Adjustments

Here are the steps for testing:

1. Connect the MOSFETs and EGS002 as per the circuit diagram.
2. Connect the 12V or 24V battery across the input.
3. Connect the output of the transformer to an AC voltmeter.
4. Power ON the circuit and measure the AC output. It should be approximately 230V for a 12V system (adjusting as required).
5. Connect a load (such as a bulb) and verify that it operates smoothly.

Step 7: Advantages of Pure Sine Wave Inverters



Compatible with all sensitive electronics.

Reduced heating and noise in motors and electronics.

Provides clean and stable AC output.

Longer equipment lifespan.

Step 8: Video Reference



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Step 9: Conclusion

Building a Pure Sine Wave Inverter using an EGS002 module and a UPS Transformer is an ideal project for electronics enthusiasts. The EGS002 greatly simplifies design, making it one of the best options for making a high-performance inverter.

With this guide and the **circuit diagram**, you can build a robust and reliable inverter for any application. If you have questions or want help selecting the right components, feel free to ask in the comments.